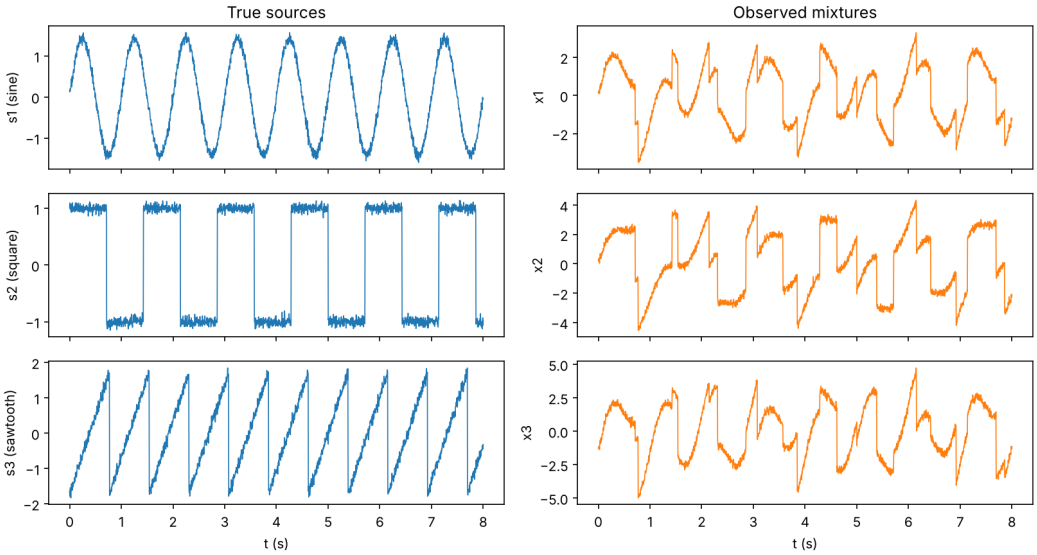
ECM-402 Lab 7: Unsupervised Learning (ICA, PCA, Clustering)

Roll no.: 523194. All results are produced by *python src/main.py* with fixed seeds (see README). **Label statement:** in every part, ground-truth labels were used only for colouring plots and for external evaluation (ARI, NMI, contingency tables). They were never passed to fit() and never used to choose k or the number of components. Dataset: synthetic modulation dataset (20 dB, seed 0, 1000 samples, 12 features).

Part A - ICA and blind source separation

A1. Sources and mixtures



(a) Every mixture is a weighted sum of all three sources, so none of the observed channels looks like a single source; the sine, square and sawtooth shapes are superimposed with different weights and cannot be unscrambled by eye, because separating them needs a statistical criterion (independence / non-Gaussianity).

(b) cond(A) = 9.854 (det A = 1.000). Moderate: A is well invertible. As cond(A) grows, columns (or rows) become nearly linearly dependent, the mixtures carry almost the same information, and A^{-1} amplifies noise/estimation error by up to cond(A); separation becomes ill-conditioned and the recovered sources degrade.

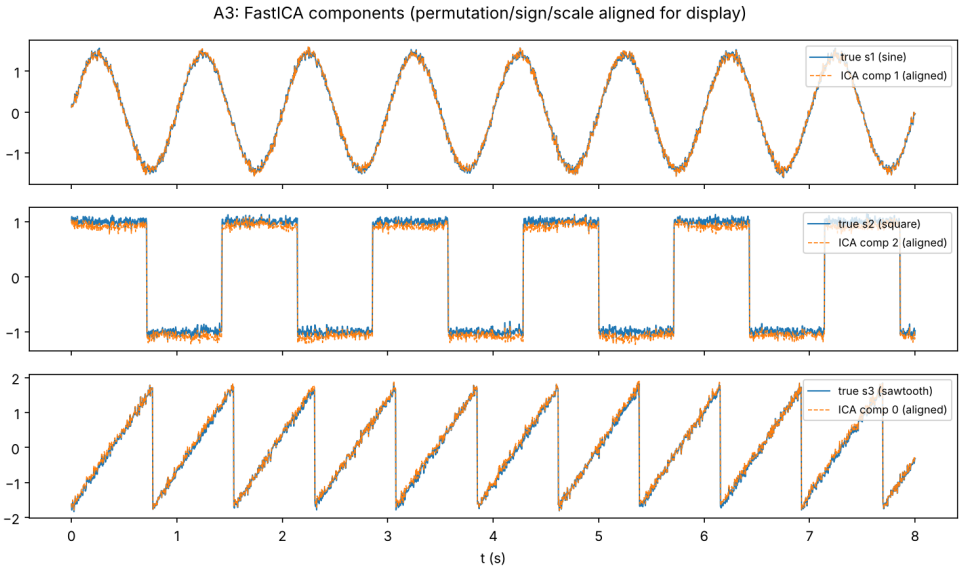
A2. Centering and whitening

cov(X_w) (ddof = 1 throughout):

1.0000	-0.0000	-0.0000
-0.0000	1.0000	-0.0000
-0.0000	-0.0000	1.0000

Max |cov(X_w) - I| = 1.3e-14, so the whitened data has identity covariance. (b) Whitening uses second-order statistics to fix n(n+1)/2 = 6 of the 9 entries of the 3x3 unmixing matrix; what is left is an orthogonal matrix with n(n-1)/2 = 3 free parameters. ICA only has to search a 3-parameter rotation instead of a 9-parameter general matrix, which makes the optimisation smaller, better conditioned, and lets it be posed as a constrained rotation problem on non-Gaussianity.

A3. FastICA



Matrix of |Pearson correlation| (rows: true sources, columns: ICA components):

	ICA0	ICA1	ICA2
sine	0.0001	0.9998	0.0223
square	0.0148	0.0182	0.9997
sawtooth	0.9996	0.0295	0.0011

One-to-one matching (Hungarian algorithm on |corr|): sine->ICA1, square->ICA2, sawtooth->ICA0; matched |corr| = 0.9998, 0.9997, 0.9996; **mean = 0.9997**. FastICA converged in 7 iterations with no warnings.

(c) The tiny deviation from 1 (about 3e-4) is consistent with the added source noise (std 0.05), finite N = 2000, and the fact that deterministic periodic waveforms are only approximately independent; no threshold is implied. (d) ICA can return any scale and sign (e.g. -3 s_i), so MSE against s_i would be large even for perfect separation, while |Pearson correlation| is invariant to scale, sign and offset, and combined with the matching handles permutation as well.

A4. PCA vs ICA

(a) PCA mean matched |corr| = **0.7023** (per source: 0.6200, 0.6708, 0.8159) versus ICA 0.9997. PCA returns orthogonal directions of maximal variance. The source directions (columns of A) are not orthogonal, so an orthonormal basis cannot align with them; and any rotation of whitened data is also uncorrelated, so covariance alone cannot pick the right one.

(b) Sample covariances:

PCA outputs			ICA outputs		
10.2419	-0.0000	-0.0000	1.0005	0.0000	0.0000
-0.0000	0.9152	-0.0000	0.0000	1.0005	-0.0000
-0.0000	-0.0000	0.1066	0.0000	-0.0000	1.0005

Both are diagonal (PCA variances 10.24, 0.92, 0.11; ICA identity), so both achieve decorrelation. This shows only that there is no *linear* dependence; it says nothing about higher-order dependence.

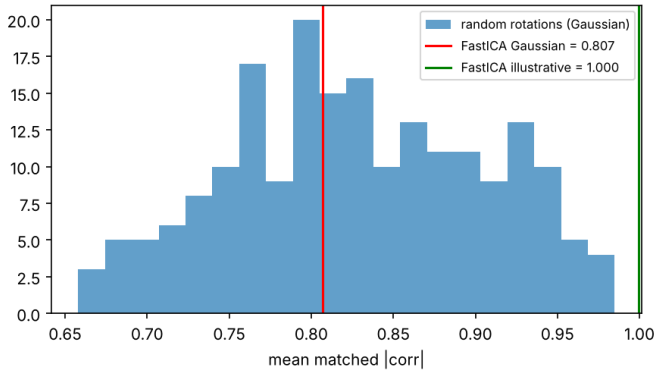
(c) $D_{ij} = E[z_i^2 z_j^2] - E[z_i^2]E[z_j^2]$ computed on centred, unit-variance outputs (so PCA and ICA are on a common scale):

D (PCA)			D (ICA)		
0.908	-0.573	-0.265	0.811	0.014	0.008
-0.573	1.710	-0.510	0.014	0.523	0.015
-0.265	-0.510	1.522	0.008	0.015	0.034

Mean |off-diagonal D|: PCA = 0.4496, ICA = 0.0124. PCA's components, though uncorrelated, have clearly non-zero correlation between their squares (energies), i.e. they are dependent; ICA's off-diagonals are about 30-40x smaller and close to zero, consistent with (though not proving) independence. Diagonals are variances of squared components and are not expected to be zero.

(d) PCA enforces only second-order decorrelation; independence requires the joint distribution to factorise, which is much stronger. (b) shows both outputs are uncorrelated, yet (c) exposes dependence in PCA's output that ICA removes. FastICA does not minimise Eq. (5); it maximises non-Gaussianity of the components.

A5. Gaussian sources: unidentifiable



(a) Random-rotation baseline (200 Haar rotations of whitened X_{gauss}): mean = 0.8274, 5th percentile = 0.7010, 95th percentile = 0.9509. (Chance level is far from 0 because the matching picks the largest entries.)

(b) FastICA on Gaussian mixtures: mean matched |corr| = **0.8068**, inside the baseline 5-95% range and not distinguishable from an arbitrary rotation. FastICA on the illustrative non-Gaussian signals (A3): 0.9997, far above the range. Convergence warnings (Gaussian case): none. Note that absence of a warning does not mean success here.

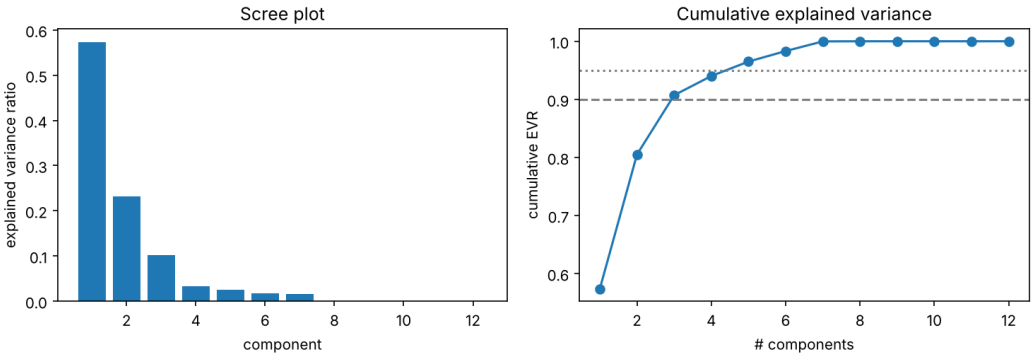
(c) For i.i.d. Gaussian sources, whitened data is N(0, I), and any rotation Q of it is again N(0, I): every orthogonal Q gives an equally valid set of independent unit-variance components. Since the distribution is rotation-invariant and second-order statistics are all that exist, nothing in the data selects the true rotation, so the sources are unidentifiable (the standard condition allows at most one Gaussian source).

(d) Excess kurtosis (Fisher): illustrative sources = -1.479, -1.968, -1.191 (theory: sine -1.5, square -2, sawtooth -1.2); Gaussian sources = -0.046, 0.082, 0.111 (theory 0). Non-zero kurtosis (non-Gaussianity) breaks the rotation symmetry: only the true rotation maximises the non-Gaussianity of the outputs, which FastICA exploits. Sample kurtosis of Gaussian data fluctuates by about sqrt(24/N) = 0.11 at N = 2000, and the sources include added noise and are standardised on finite samples.

Part B - PCA for dimensionality reduction (modulation dataset, 20 dB)

B6. Standardisation and intrinsic dimensionality

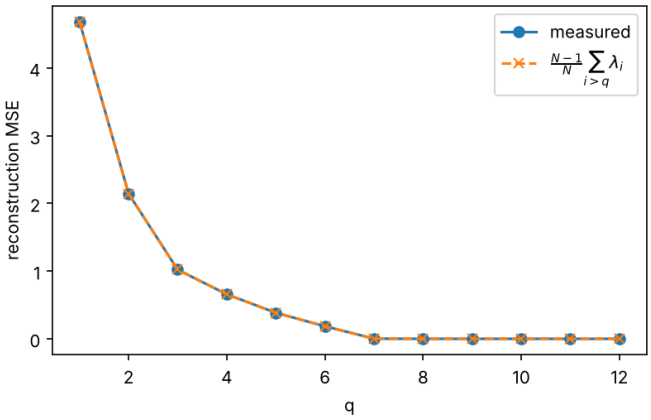
(a) Standardisation is needed because PCA maximises variance, so features with large numerical scale (e.g. kurtosis-type features) would dominate the components purely due to units.



(b) Explained variance ratio of PC1-PC7: 0.574, 0.231, 0.102, 0.033, 0.025, 0.018, 0.017; PC8 = 0.0002, PC9-12 are about 0. Components needed: **3 for 90%**, **5 for 95%** variance.

(c) Not necessarily equal to four, and there is no reason to expect it. Intrinsic dimensionality measures how many independent directions of variation the features have (here only ~8 of 12 carry variance, because several features are exactly redundant after unit-power normalisation, e.g. |C40| and |C40|/C21^2, and mean|r|^2 is constant), including within-class spread from noise and carrier phase. The number of classes is a label count: the four class centroids span at most a 3-D affine subspace, but the variance needed to reach 90% depends on the feature correlations and noise, not on the number of classes. (Getting 3 here is a coincidence of this dataset.)

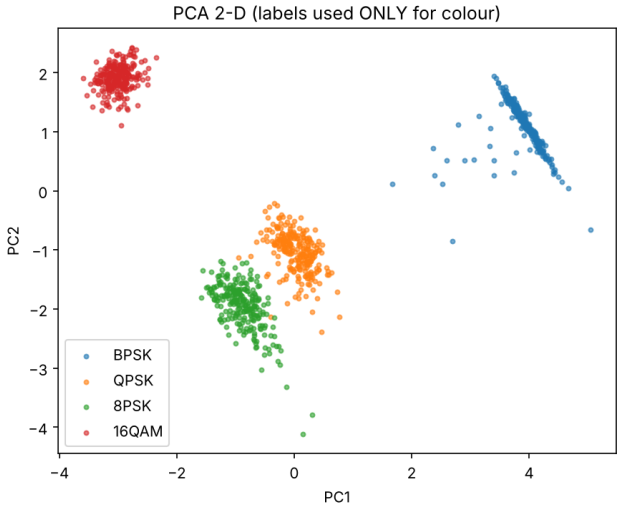
B7. Reconstruction error



q	MSE measured (Eq. 6)	(N-1)/N * sum of discarded lambda (Eq. 7)
1	4.688053	4.688053
3	1.021504	1.021504
5	0.384658	0.384658
7	0.002401	0.002401

Measured and predicted values agree to machine precision for every q (full table in results/tables/B7_reconstruction_mse.csv). The error left by a rank-q projection is exactly the variance in the discarded directions; since PCA picks the directions with the largest eigenvalues, it leaves the smallest possible discarded sum, which is the least-squares (Eckart-Young) optimality of PCA.

B8. Two-component visualisation



(a) **Labels were used only for colouring this plot; PCA was fitted on the standardized features alone.** PC1+PC2 retain 80.5% of the variance.

(b) BPSK and 16QAM are clearly isolated; **QPSK and 8PSK are the closest pair** and touch at the boundary (centroid distance 3.9 within-class std units vs 12.6+ for all other pairs). A 2-D projection keeps only ~80% of the variance; the discarded directions can contain exactly the small differences (e.g. higher-order cumulant combinations) that separate similar classes.

(c) Largest PC1 loadings: |C42| (0.386), |C42|/C21^2 (0.386), |C20| (0.352) (|C42| and $|C42|/C21^2$ are identical features here). These are fourth-order and second-order cumulant magnitudes. They make physical sense: C40 and C42 are amplitude/phase-structure statistics that differ sharply between PSK (theoretical |C42| = 2 for BPSK, 1 for QPSK/8PSK) and QAM (non-constant amplitude, |C42| = 0.68), so they carry most of the PSK-vs-QAM and BPSK-vs-others variance.

Part C - k-means

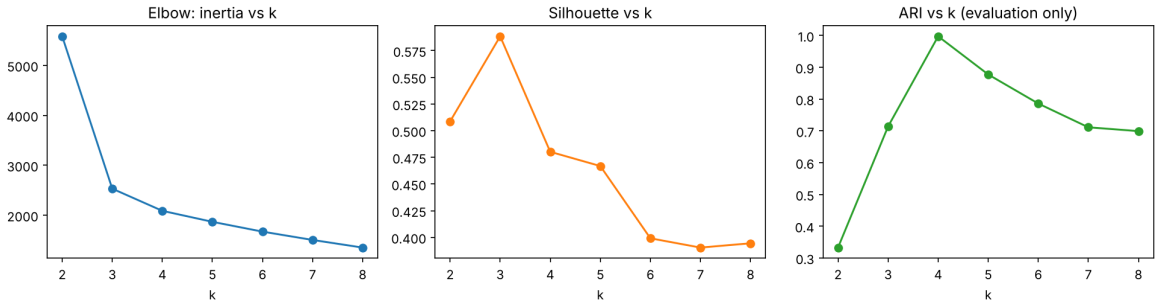
C9. k = 4 on the 12-D standardized features

(a) Inertia = 2088.20, silhouette = 0.4802. (b) ARI = 0.9973, NMI = 0.9953 (labels used only for this evaluation).

true \ cluster	0	1	2	3
BPSK	0	249	0	1
QPSK	0	0	0	250
8PSK	250	0	0	0
16QAM	0	0	250	0

(c) Cluster indices are arbitrary names (a permutation of them gives the same clustering), so plain accuracy that compares cluster id with class id is meaningless without first finding the optimal alignment. (d) ARI is the Rand index corrected for chance: the raw Rand index of random partitions is well above 0, so ARI subtracts its expected value under random labelling and rescales to 1 for a perfect match; random assignments give ARI approx. 0 on average (500 random 4-way assignments: mean -0.00008, std 0.0014), but any single random assignment is only close to 0 and can be slightly negative or positive because of sampling variability.

C10. Choosing k without labels



k	inertia	silhouette	ARI (eval only)
2	5581.4	0.5087	0.3327
3	2531.9	0.5884	0.7137
4	2088.2	0.4802	0.9973
5	1868.4	0.4668	0.8772
6	1672.0	0.3991	0.7858
7	1505.7	0.3904	0.7111
8	1352.7	0.3944	0.6991

(b) Using internal criteria only: inertia falls steeply to k = 3-4 and then flattens gradually (an elbow at about 3-4 that is not sharp), and silhouette is maximal at k = 3 (0.588 vs 0.480 at k = 4). The silhouette criterion therefore selects **k = 3**; the elbow alone would be ambiguous between 3 and 4, and I would state that honestly.

(c) Highest silhouette: k = 3. Highest ARI: k = 4 (ARI 0.997 at k = 4 vs 0.714 at k = 3). They disagree. QPSK and 8PSK lie close together (Part B8), so merging them gives three compact, well-separated clusters and a higher silhouette even though the clusters no longer match the true classes. Silhouette measures compactness vs separation, not agreement with the semantic classes.

(d) Without labels, a single internal index can select a coarser (or finer) partition than the real structure. Safeguards: use several indices (elbow, silhouette, Davies-Bouldin/Calinski-Harabasz, gap statistic), check cluster stability under resampling/seeds, and use domain knowledge (here: four known modulation types).

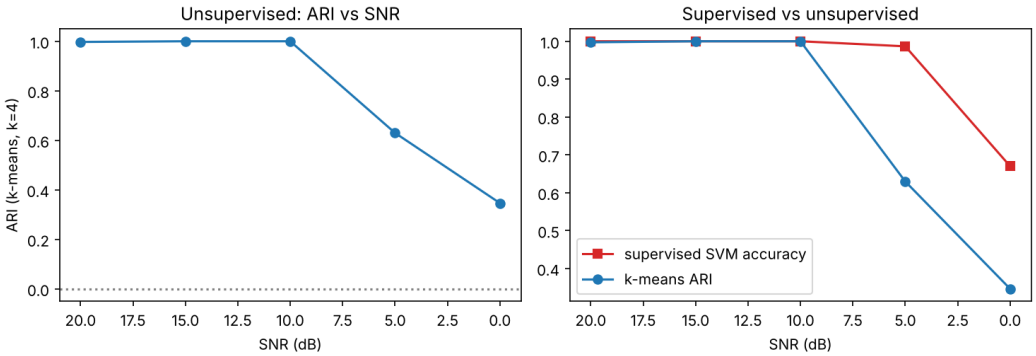
C11. Does PCA before clustering help?

representation	silhouette	ARI	NMI

12-D standardized	0.4802	0.9973	0.9953
2-D PCA	0.7439	0.9867	0.9805

Reducing 12-D to 2-D did **not** improve ARI (0.997 to 0.987, slightly lower); the silhouette rose from 0.48 to 0.74. In 2-D the discarded dimensions (noise-dominated and redundant features) no longer add intra-cluster spread, and distances in low dimensions are less diluted, so clusters look more compact and separated; but the projection also throws away information, which is why ARI does not improve (a few more samples are misassigned). Silhouette scores computed in different feature spaces are therefore not comparable: they measure geometry of that space, not clustering correctness.

C12. Clustering vs SNR



SNR (dB)	k-means ARI	NMI	silhouette	supervised SVM acc.
20	0.9973	0.9953	0.4802	1.0000
15	1.0000	1.0000	0.4503	1.0000
10	1.0000	1.0000	0.3871	1.0000
5	0.6297	0.7709	0.3519	0.9867
0	0.3462	0.4092	0.2049	0.6700

(a,b) ARI is essentially perfect from 20 to 10 dB, drops sharply to 0.63 at 5 dB and 0.35 at 0 dB. Random clustering would give ARI approx. 0; even at 0 dB the result (0.35) remains well above this baseline, so the clusters still contain real information. (c) The Lab-3 curve was not available to me; as a stand-in I trained an RBF-SVM on the same features (70/30 split, seed 0). It keeps 0.987 accuracy at 5 dB and 0.67 at 0 dB, whereas k-means starts to fail at 5 dB (ARI 0.63). Both methods work down to 10 dB; unsupervised clustering degrades first because with noise the clusters spread and overlap and k-means must find the boundaries from geometry alone, whereas a supervised classifier is told where the boundaries lie and can learn non-spherical, noise-aware decision regions. (Accuracy and ARI are different scales, so the comparison is indicative.)

Part D - When k-means is the wrong tool

D13. KMeans vs GMM vs Agglomerative (20 dB, k = 4)

method	ARI	NMI
KMeans	0.9973	0.9953
GMM (full)	1.0000	1.0000
Agglomerative (ward)	1.0000	1.0000

Seeds: dataset seed 0; KMeans random_state 0 (n_init = 10); GMM covariance_type = 'full', random_state 0, n_init = 5; Agglomerative linkage = 'ward'.

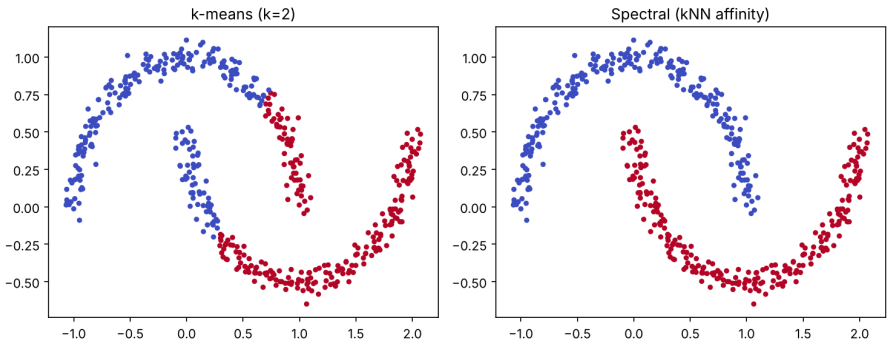
Additional check: the same comparison repeated on 10 independent datasets (seeds 100-109):

method	ARI mean +/- std	NMI mean +/- std
KMeans	1.0000 +/- 0.0000	1.0000 +/- 0.0000

GMM (full)	0.9685 +/- 0.0945	0.9777 +/- 0.0670
Agglomerative (ward)	0.9822 +/- 0.0489	0.9862 +/- 0.0349

(a) In the single run GMM and Agglomerative reach ARI = NMI = 1.0, ahead of k-means (0.997, one misassigned BPSK sample); but this difference is one sample. In the 10-dataset repeat the ordering reverses (k-means 1.000 +/- 0, GMM 0.969 +/- 0.095, Agglomerative 0.982 +/- 0.049), showing that a single run cannot establish that one algorithm is better: results depend on the random dataset and initialisation (GMM fits full 12-D covariances on near-degenerate features, which sometimes gives unstable fits). (b) A GMM relaxes k-means' assumption by giving each cluster its own full covariance matrix (elliptical, differently oriented/sized clusters) and soft assignment probabilities, rather than spherical equal-variance clusters.

D14. make_moons failure case



k-means ARI = 0.2566; spectral clustering ARI = 1.0000 (labels used only for ARI).

(a) k-means assigns each point to the nearest centroid, so the boundary between two centroids is a straight line (the perpendicular bisector). The two crescents are interleaved and non-convex: no straight line separates them, so k-means cuts both moons in half. (b) Spectral clustering builds a nearest-neighbour graph, so points are linked to their neighbours, and clusters are found from the eigenvectors of the graph Laplacian as groups that are well connected inside and weakly connected to each other, so each crescent (a connected chain of neighbours) becomes one cluster. Recorded warning: 'Graph is not fully connected' - here that is expected because the two moons form two disconnected components.